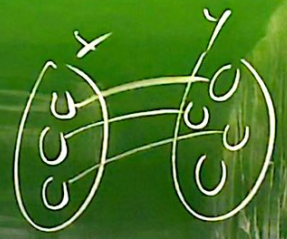
COMPARING CARDINALITIES

DEF SETS X, Y HAVE THE SAME CARDINALITY (MOHUTNOST) $X \approx Y$ IF THERE EXISTS AN INJECTIVE MAPPING OF X ONTO Y (X IS EQUIVALENT TO Y)

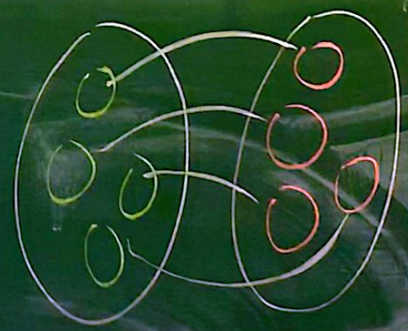
A SET X HAS CARDINALITY SMALLER THAN OR EQUAL TO THE CARDINALITY OF Y, $X \preccurlyeq Y$ (X IS SUBVALENT TO Y) IF THERE EXISTS AN INJECTIVE MAPPING FROM X TO Y



X HAS CARDINALITY SMALLER THAN Y IF $X \preccurlyeq Y \wedge \neg(X \approx Y)$

OBSERVATION $X \subseteq Y \rightarrow X \preccurlyeq Y$ (Id)

$X \subset Y \rightarrow X \preccurlyeq Y$ BUT NOT NECESSARILY $X \prec Y$



QUESTION IS $\preccurlyeq$ TRICHOTOMIC? (IN ZF IT IS UNDECIDABLE), IT IS EQUIVALENT TO (AC)

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LEMMA IF X, Y, Z ARE SETS, THEN 1) $X \approx X$ 2) $X \approx Y \to Y \approx X$ 3) $(X \approx Y \wedge Y \approx Z) \to X \approx Z$ (SO $\approx$ IS AN EQUIVALENCE)

4) $X \preceq X$ 5) $(X \preceq Y \wedge Y \preceq Z) \to X \preceq Z$

(IS $\preceq$ WEAKLY ANTISYMETRIC?

PROOF: 1) Id 2) $F: X \to Y$ F^{-1} 3) $F: X \to Y$, $G: Y \to Z$ $F \circ G: X \to Z$

4) Id 5) $F \circ G$

OBSERVATION $X \approx Y \to (X \preceq Y \wedge Y \preceq X)$

$\overset{?}{\leftarrow}$

$\{\emptyset\} \preceq \{\{\emptyset\}\}$

$\succeq$

$\approx$

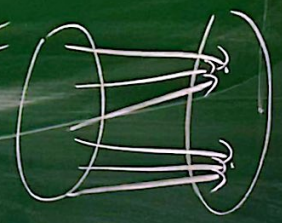
THEOREM

CANTOR-BERNSTEIN $(X \preceq Y \wedge Y \preceq X) \longrightarrow X \approx Y$

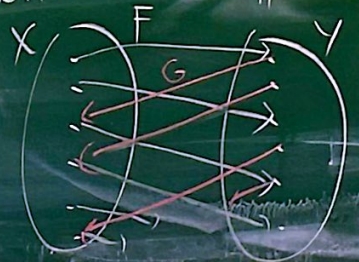
$(X \xrightarrow{\text{ONTO}} Y \wedge Y \xrightarrow{\text{ONTO}} X) \overset{?}{\to} X \approx Y$

CANNOT BE DECIDED IN ZF

(TRUE IN ZFC)



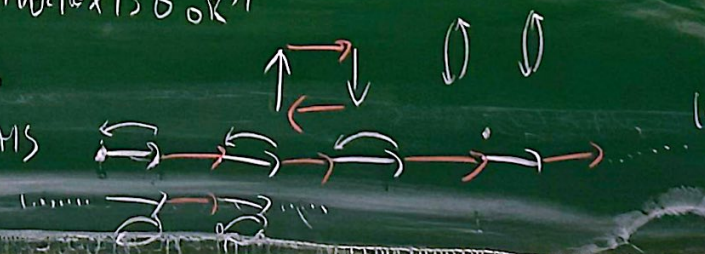
A GRAPH-THEORETIC „PROOF":



$F \cup G$ ... AN ORIENTED GRAPH WITH OUTDEGREE 1, INDEGREE OF EVERY VERTEX IS 0 OR 1

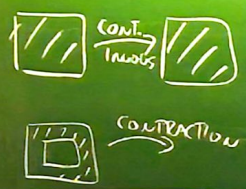
· COMPONENTS: EVEN CYCLES

INFINITE PATHS



(ODD/EVEN LENGTH)

BROUWER $\mathbb{N}\setminus\{1\}$ $\mathbb{N}$



x y

$n \to n-1$

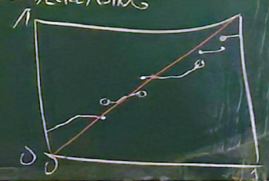
CONTRACTION

KNASTER–TARSKI (COMPLETE LATTICES)

$\wedge$ $\vee$

$\boxed{\text{EX.}}$ $h : [0,1] \to [0,1]$ NONDECREASING

THEN h HAS A FIXED POINT



PROOF USING A FIXED-POINT LEMMA:

$\boxed{\text{DEF}}$ A MAPPING $H : \mathcal{P}(X) \to \mathcal{P}(X)$ IS MONOTONE WITH RESPECT TO INCLUSION IF FOR EVERY $M, N \subseteq X$ WE HAVE $M \subseteq N \to H(M) \subseteq H(N)$

$\boxed{\text{LEMMA}}$ IF $H : \mathcal{P}(X) \to \mathcal{P}(X)$ IS MONOTONE WITH RESPECT TO INCLUSION, THEN THERE EXISTS A SET $C \subseteq X$ SUCH THAT $H(C) = C$.
(C IS A FIXED POINT OF H)

$\boxed{\text{EX}}$: $A \subseteq \mathcal{P}(X)$ THEN $\sup_{\subseteq} A = \bigcup A$ $\quad (\inf_{\subseteq} A = \bigcap A)$

PROOF: LET $A = \{M : M \subseteq X \wedge M \subseteq H(M)\}$.

LET $C = \bigcup A$ $(= \sup A)$

- $C \subseteq X$.
- LET $M \in A$ $\begin{cases} M \subseteq X \text{ THEN } M \subseteq C \quad (C \text{ IS AN UPPER BOUND OF } A) \\ M \subseteq H(M) \subseteq H(C) \quad \cdots \text{ SO } H(C) \text{ IS AN UPPER BOUND OF } A \end{cases}$ (H IS MONOTONE)

  SO $C \subseteq H(C)$ (C IS A SUPREMUM OF A) $\Big\}\ C = H(C)$
- BY MONOTONICITY OF H WE HAVE $H(C) \subseteq H(H(C))$ SO $H(C) \in A$ SO $H(C) \subseteq C$ (C IS AN UPPER BOUND)

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